Problem B

Predicting Penalty Trajectories: What Two Early Positions Can Resolve

Abstract

Two positions measured within the first tenth of a second after a penalty kick already reveal a turn toward the centre relative to the first interval's heading. For the prescribed kick, allowing centimetre rounding and a steady wind up to 3 m/s, drag alone cannot explain the turn. Our most probable goal cell is centred at $(2.56, 2.07)$ m. Conditional on direct entry, six cells spanning 0.357 m^2 contain about 90% of the model probability.

We weight trajectories by how well they reproduce the two readings, then follow each to the goal plane or the ground. Several force histories fit the observations. The inferred point lies farther from the near post than the prediction under gravity alone. The two models also respond differently to errors in the early positions, which are magnified when the flight is extrapolated to the goal.

Two admissible flights with the same displayed coordinates cross 0.92 m apart, illustrating the worst-case ambiguity hidden by rounding. In independent synthetic tests, the earlier frame narrowed the forecast region under rounding alone. With added tracking error, the aerodynamic regions covered 93% of direct entries overall and 82% of stable-spin entries. Cleaner measurements improved point accuracy. Increasing the region cutoff recovered misses at the cost of a larger area. Three match records failed the two-frame launch checks, leaving accuracy on real penalties unmeasured.

1 What the first 100 ms reveal about the flight

Problem restatement. A goalkeeper has two positions of the ball centre, measured 50 and 100 ms after a penalty leaves the mark. We seek the most likely point of direct entry through the goal plane and the uncertainty in that point, using only information available at the second reading [1]. We treat contact with the ground before the plane as a different outcome. For the prescribed kick, our highest-probability cell is centred at $(Y, Z) = (2.56, 2.07)$ m, about a metre inside the positive-y post.

By the second reading the ball has travelled 2.96 m toward the goal and has 8.04 m still to fly (Fig. 1). Its horizontal displacements show that the heading is turning toward smaller y relative to the first 50 ms chord. We first measure that turn, then ask which force histories could produce it and where their paths cross the goal plane. The prescribed kick has no observed endpoint, so we test the method on separate kicks (Table 1).

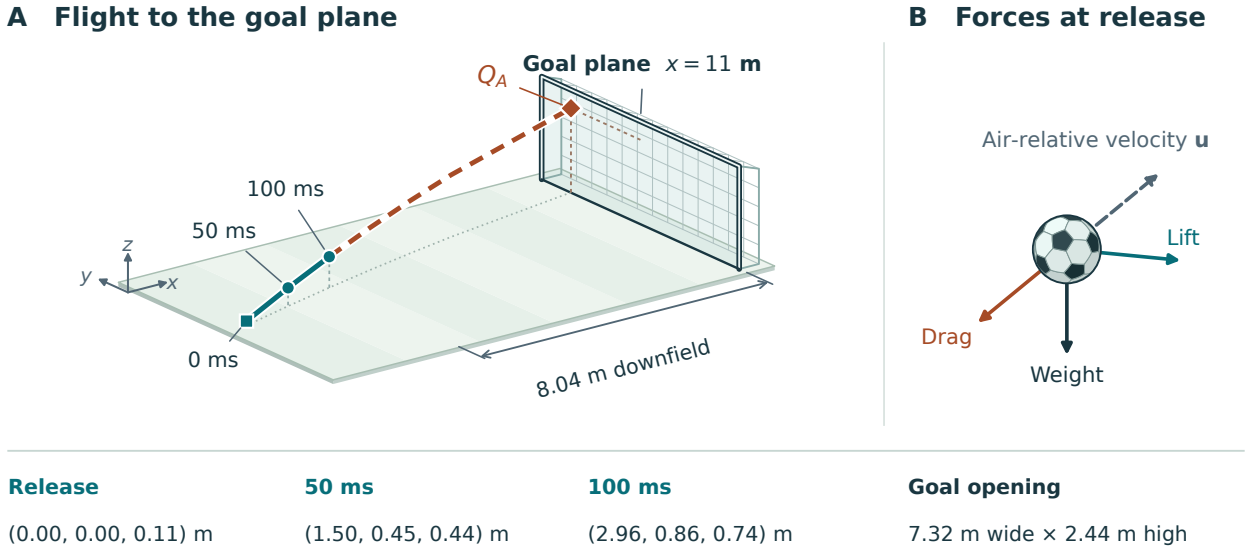


Figure 1. Two readings and one admissible flight. In A, the square marks release and the circles mark the prescribed ball centres, given below in (x, y, z) order. Dashed segments join archived positions of flight A to its crossing Q_A (Sec. 4). The dotted line on the turf is the ground projection. B shows the same flight at release in still air. Drag opposes the air-relative velocity, lift acts perpendicular to it in three dimensions, and weight points down. Arrow lengths are schematic.

Table 1. How the two observations available by 100 ms enter each part of the problem.

Task	Forecast or test	Evidence
Develop a predictor	First-event probabilities and highest-mass goal cell	Eqs. (3), (5)
Estimate uncertainty	Joint region and sensitivity to observation and force assumptions	Figs. 3–5
Apply and test	Prescribed kick and synthetic kicks at the same 100 ms cutoff	Eq. (6), Fig. 6

1.1 Observations and possible outcomes

Let x point from the penalty spot toward the goal, y run across its face and z measure height above the ground. The ball centre starts at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m. Its two prescribed positions, in metres, are

$$\boldsymbol{r}(0.050\,\mathrm{s}) = (1.50, 0.45, 0.44), \qquad \boldsymbol{r}(0.100\,\mathrm{s}) = (2.96, 0.86, 0.74). \tag{1}$$

For the reference forecast we take the release position and observation times as exact and interpret the displayed coordinates as rounded to the nearest centimetre. Each later coordinate q then represents $[q - \epsilon, q + \epsilon)$, with $\epsilon = 0.005$ m. The problem supplies no tracking accuracy, ball type, wind or spin. Section 5 examines how different observation assumptions affect this kick.

We follow the ball to its first forward crossing of $x = 11$ m or its first descent to one ball radius above ground. A direct entry G occurs when the centre crosses the 7.32 m wide opening with $-3.66 < Y < 3.66$ m and $0 < Z < 2.44$ m. The other outcomes are a crossing outside the opening, earlier ground contact, or neither event by 1.25 s. We estimate all four probabilities and the distribution of $Q = (Y, Z)$ conditional on G. This ball-centre convention omits frame collisions, rebounds and goalkeeper interventions.

1.2 How much wind would drag alone require?

The mean horizontal velocities over the first and second 50 ms are $(30.0, 9.0)$ and $(29.2, 8.2)$ m/s. Their directions differ. Under drag alone in a constant horizontal wind $\boldsymbol{w}$, the horizontal velocity keeps its direction relative to the air. The horizontal displacements $\boldsymbol{p}_i$ from release must therefore satisfy

$$\det(\boldsymbol{p}_1 - t_1\boldsymbol{w},\, \boldsymbol{p}_2 - t_2\boldsymbol{w}) = 0, \qquad \|\boldsymbol{w}\| \geq \frac{|\det(\boldsymbol{p}_1, \boldsymbol{p}_2)|}{\|t_2\boldsymbol{p}_1 - t_1\boldsymbol{p}_2\|}. \tag{2}$$

At the displayed coordinates, Eq. (2) requires at least 14.85 m/s. To allow for rounding, fix the wind first. The determinant is affine in each coordinate separately, so its value at any point of the four-dimensional box is a weighted average of its values at the 16 corners. All corners have the same sign at zero wind. The first possible zero as the wind disk expands is therefore a corner zero. Evaluating Eq. (2) at the corners gives an infimum of 4.353 m/s for the rounding intervals, above the assumed cap of 3 m/s. Collinearity is a necessary condition for a drag-only flight.

How large is the turn? With $h = 0.050$ s, the horizontal second difference $(\boldsymbol{p}_2 - 2\boldsymbol{p}_1)/h^2 = (-16, -16)$ m/s^2 is the net acceleration averaged over the first $2h$ with a triangular time weight. Projecting onto the left normal of the full air-relative chord gives -10.9 m/s^2 in still air at the displayed coordinates. The smallest magnitude over the allowed winds is about 8.92 m/s^2 at those coordinates. Allowing all four horizontal coordinates to vary within their rounding intervals lowers the infimum to about 1.07 m/s^2. Under our force laws, the horizontal component of non-drag acceleration toward the inside of that turn must reach at least 0.97 m/s^2 at some instant (Appendix C.2). These observations resolve a turn while leaving its cause and the spin rate uncertain.

In still air the horizontal path under drag alone must be a straight line through release. The slope intervals $[(y_i - \epsilon)/(x_i + \epsilon), (y_i + \epsilon)/(x_i - \epsilon)]$ first touch at $\epsilon = 7.279$ mm for this kick. A 5 mm per-axis bound resolves the turn, whereas a 1 cm bound admits a straight path.

2 From aerodynamic forces to a crossing forecast

2.1 Three force laws and their parameter ranges

Write $\boldsymbol{v} = \dot{\boldsymbol{r}}$ for velocity over the ground, $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{w}$ for velocity relative to the air, and $U = \|\boldsymbol{u}\|$. The three force models give accelerations

$$\begin{aligned} \boldsymbol{a}_{\mathrm{D}} &= \boldsymbol{g} - \kappa U \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{S}} &= \boldsymbol{a}_{\mathrm{D}} + \boldsymbol{\beta} \times \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{E}} &= \boldsymbol{g} - K C_D U \boldsymbol{u} + K C_L U^2 \boldsymbol{n}. \end{aligned} \tag{3}$$

Model D has quadratic drag. S adds a transverse force with constant coefficient vector $\boldsymbol{\beta}$. In E, $K = \rho\pi R^2/(2m)$ and lift acts along the unit vector $\boldsymbol{n}$ in the direction of $\boldsymbol{\omega} \times \boldsymbol{u}$. Lift vanishes when the cross product is zero. We use $g = 9.81$ m/s^2 downward, radius $R = 0.11$ m, mass $m = 0.43$ kg and air density $\rho = 1.20$ kg/m^3. With inputs in SI units, the coefficients for E are

$$\begin{aligned} C_D &= 0.155 + \frac{0.346}{1 + \exp[(U_c - 12.19)/1.309]} + \delta, \qquad U_c = \min(29.9, \max(9.3, U)), \\ C_L &= 0.38 \tanh\!\left(5R\|\boldsymbol{\omega} \times \boldsymbol{u}\|/U^2\right). \end{aligned} \tag{4}$$

For the speed-dependent part of C_D, Goff and Carré [2] fitted their Eq. (3) and Fig. 1 to the nonspinning Adidas Teamgeist data of Asai et al. [3], whose tunnel tests of nonrotating balls covered 7–35 m/s. We clip the speed argument to 9.3–29.9 m/s using the initial-speed range of a separate 29-flight study [4]. The offset δ, the saturating lift law and the spin limits are modelling choices for the unspecified ball. In S, $\boldsymbol{\beta}$ represents an effective transverse-force coefficient.

We assign D, S and E prior probabilities $0.2, 0.4, 0.4$ and draw wind uniformly from a horizontal disk of radius 3 m/s. Launch-velocity components are independently uniform over $[18, 38]$, $[-12, 12]$ and $[1.5, 11]$ m/s along x, y, z. For D and S, κ is uniform on $[0.006, 0.024]$ m^{-1}. The components of $\boldsymbol{\beta}$ in S and $\boldsymbol{\omega}$ in E are uniform within ± 0.8 s^{-1} and ± 65 rad/s, respectively. In E, δ is uniform on $[-0.04, 0.27]$.

2.2 Force scales over the remaining flight

The first displacement gives a speed scale $U_0 \simeq 32$ m/s. At $\kappa \simeq 0.020$ m^{-1}, the 0.4 s flight lasts about 0.26 of the drag time $(\kappa U_0)^{-1} \simeq 1.56$ s. A transverse coefficient of 0.5 s^{-1} gives an initial acceleration scale of 16 m/s^2. A residual transverse acceleration of 1 m/s^2 sustained over the remaining 0.30 s would accumulate about 4.5 cm of displacement.

Constant wind also gives an energy check. Because $\boldsymbol{u} \cdot (\boldsymbol{\beta} \times \boldsymbol{u}) = 0$ and $\boldsymbol{u} \cdot \boldsymbol{n} = 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{U^2}{2} = \boldsymbol{u} \cdot \boldsymbol{g} - d(U)U^2, \qquad d(U) \geq 0,$$

where $d(U)$ is the nonnegative drag rate. A transverse force turns the air-relative velocity, while drag dissipates air-relative kinetic energy. We use this identity to bound acceleration in the match records.

2.3 Weighting trajectories and mapping their crossings

For each draw of wind and aerodynamic parameters, we estimate launch velocity from the latest visible position under gravity and correct the estimate three times by integrating to that position. We draw velocities from a local uniform distribution centred on this estimate, with half-width $0.007\,\mathrm{m}/t$ per axis at observation time t, or from the original prior with probability 2%. Under rounding alone, a trajectory inside every observed interval receives the prior-to-proposal density ratio. All others receive zero weight. For parameters θ, proposal density q_M and likelihood L,

$$w_{Mj} = \frac{p(\theta_{Mj} \mid M)L(D \mid \theta_{Mj}, M)}{q_M(\theta_{Mj})}, \quad \widehat{Z}_M = \frac{1}{N_M}\sum_j w_{Mj}, \quad \widehat{p}(M \mid D) \propto p(M)\widehat{Z}_M. \tag{5}$$

Sampling nuisance parameters from their priors cancels those densities in the ratio. The correction for launch velocity remains. The mean weight $\widehat{Z}_M$ estimates the evidence for force model M. Systematic resampling limits propagation to 512 trajectories per model per test kick, or 8,192 per model per seed for the prescribed kick. Both prescribed-case runs stayed below their cap.

We use fourth-order Runge–Kutta steps of 25 ms through the two prescribed observations and at most 12.5 ms thereafter. Linear interpolation locates plane crossings and ground contact within a step. For the four flights in the original solver check, independent integrations including SciPy DOP853 [5] agreed within 0.171 mm in event position and 0.038 ms in time.

First-event probabilities use the weights of all trajectories. For direct entries we normalize their weights on a 30×10 equal-area goal grid. The centre of the most probable cell gives the point forecast. Adding cells in decreasing order of probability until they contain at least 90% gives the raw joint region (Fig. 2). Zero total weight leaves the forecast undefined. If only the direct-entry weight is zero, we can still report event probabilities. We flag test kicks with an effective sample size $\mathrm{ESS} = 1/\sum_j \bar{w}_j^2$ below 25, using normalized weights including the model and timing mixtures. For the more densely sampled prescribed kick we flag pooled ESS below 200 and require ESS of at least 25 in any model carrying at least 10% of the posterior mass. These prespecified numerical checks flag weight concentration. They provide no confidence level for the physical accuracy of a forecast.

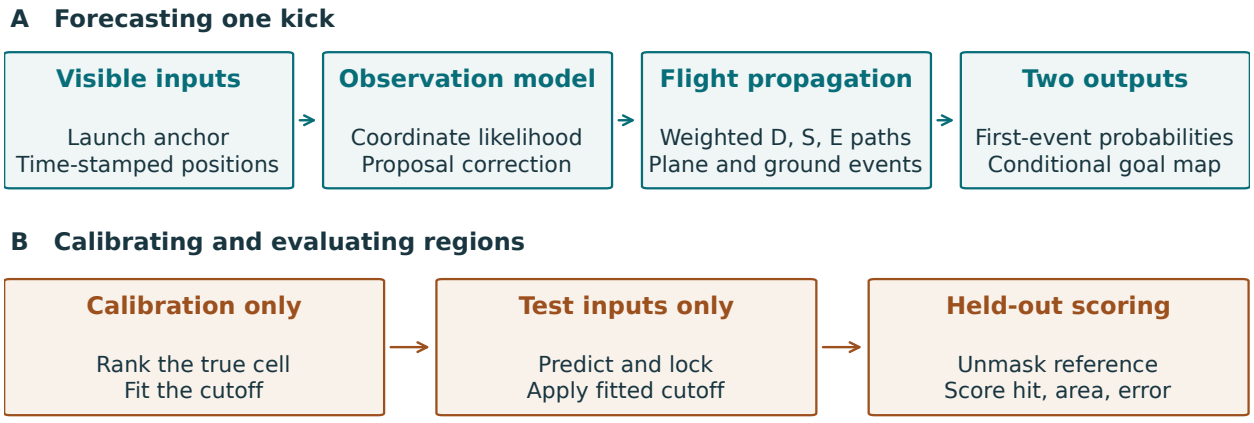


Figure 2. From early positions to a forecast. A follows one kick from its observed positions to first-event probabilities and a location map conditional on direct entry. B shows how we calibrate and test regions on separate kicks (Sec. 6). Only scoring uses the held-out reference crossings.

3 Where the prescribed kick is most likely to cross

3.1 Modal cell and joint prediction region

Conditional on direct entry, the most probable cell contains 29.1% of the probability (Fig. 3). Its centre, approximately $(Y, Z) = (2.56, 2.07)$ m, lies 1.098 m inside the positive-y post and 0.366 m below the crossbar. The six most probable cells form the rectangle

$$2.196 \le Y \le 2.928 \text{ m}, \qquad 1.708 \le Z \le 2.196 \text{ m}. \tag{6}$$

Together these cells cover 0.357 m^2 and contain 90.5% of the conditional probability. These figures summarize masses on the specified grid. The median crossing time from release is 0.399 s, with a central 90% interval of 0.392–0.407 s conditional on exact frame times. Section 5 examines uncertain frame timing.

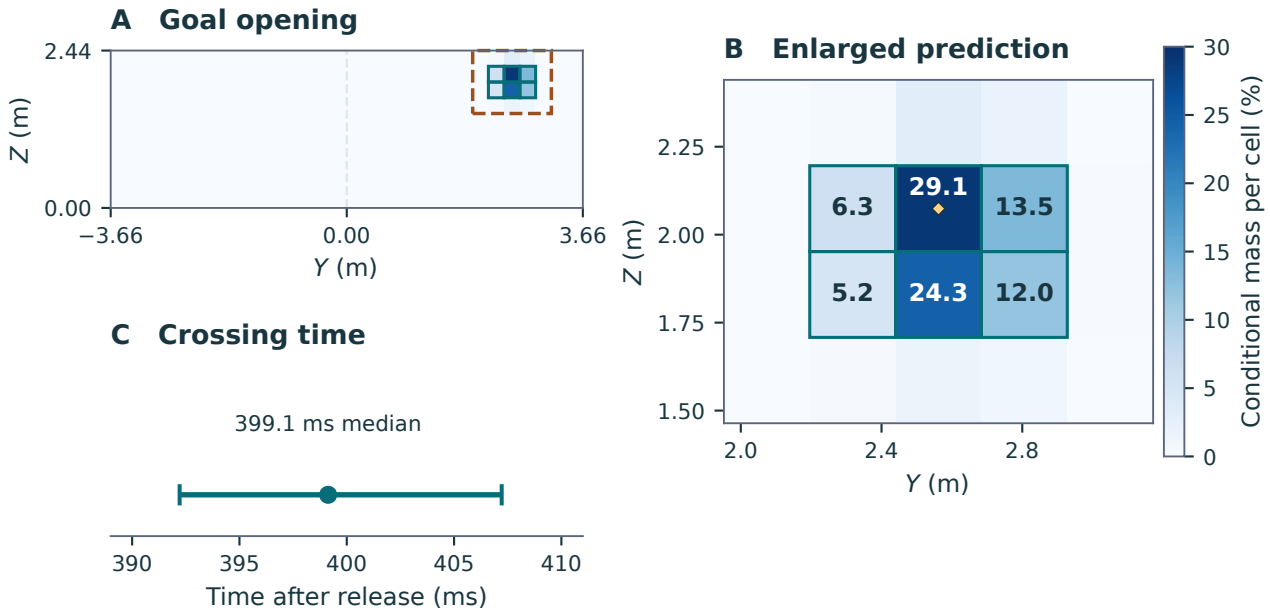


Figure 3. Predicted location and time of direct entry. A locates the enlarged area on the goal. B shows the probability per 0.244 m square cell using the same colour scale. The outline encloses the six most probable cells and the diamond marks the modal centre. C gives the 5th, 50th and 95th percentiles of the crossing time.

All 3,469 positive-weight trajectories from two independent runs of 131,072 proposals per model reached the opening, giving a finite-sample estimate $\widehat{P}(G) = 1$. A rare miss could remain unsampled. The model-stratified importance weights do not give a binomial confidence bound on the population probability. Posterior model weights were $0, 0.346, 0.654$ for D, S and E. The zero weight for D also follows from the wind bound in Eq. (2). The crossing of the prescribed kick has no measured reference.

3.2 Which physical parameters do the readings constrain?

Replaying the same two runs gives the distributions in Table 2. In model E, the median release lift coefficient shifts from 0.277 under the prior to 0.220 given the readings. The spin-speed interval remains broad and the wind-speed distribution changes little. Only 6.3% of E's posterior weight puts any spin component in the outermost 2% at either face of its allowed interval, compared with 11.5%

under the prior. The fit gives little support for a spin rate near the imposed limit. An independent spin measurement would distinguish among the remaining explanations.

Table 2. Parameters of the prescribed kick. Each entry gives the median [5th, 95th percentile] within the specified force model. The last column compares posterior and prior mass in the top 2% of the scalar range. For C_L the permitted launch and spin ranges give a maximum of 0.37958 at release. For spin speed it is the corner of the component cube, $65\sqrt{3}$ rad/s. Zero denotes no sampled mass in the band.

Model	Quantity	Prior	After both readings	Upper band (%)
S	κ (m^{-1})	0.0150 [0.0069, 0.0231]	0.0198 [0.0149, 0.0233]	2.83 / 2.00
E	C_L (release)	0.277 [0.107, 0.356]	0.220 [0.134, 0.301]	0 / 0.38
E	Spin speed (rad/s)	64.0 [29.7, 89.8]	55.9 [29.9, 81.3]	0 / 0.004
S	Wind speed (m/s)	2.12 [0.67, 2.92]	2.03 [0.61, 2.90]	3.00 / 3.96
E	Wind speed (m/s)	2.12 [0.67, 2.92]	2.08 [0.66, 2.91]	3.31 / 3.96

3.3 How precisely can the hotspot be located?

Both seeds selected the cell centred at $(2.562, 2.074)$ m. Varying E's share of the combined S/E prior from 10% to 90% also preserved this cell. A finer grid moved the modal centre to $(2.501, 2.013)$ m, and a half-cell shift of the original grid gave $(2.684, 1.952)$ m. The extra digits identify a cell on each grid. The six-cell region in Eq. (6) quantifies the nominal spatial uncertainty. The continuous conditional mean, $(2.603, 1.970)$ m, minimizes posterior squared distance. Our aerodynamic point forecasts use the centre of the cell with greatest probability at the stated resolution.

Gravity alone, with launch velocity fitted by least squares, predicts $(3.217, 1.957)$ m, just 0.44 m inside the near post. The aerodynamic modal cell lies 0.65 m farther toward the centre. This difference measures the combined effect of refitting the launch state and changing the force model and point estimator. A constant-acceleration fit through all three positions gives $(2.495, 1.920)$ m.

4 A lower bound on two-position prediction error

4.1 Two flights consistent with the same observations

How closely can any point forecast fit every admissible flight? In model S we fixed the wind at zero and β_x at either -0.5 or $+0.5$ s^{-1}. For each choice we fitted the six coordinates at 50 and 100 ms by varying the three launch-velocity components and κ, β_y, β_z. Both fits lie within the prior ranges, with maximum coordinate residual below 5×10^{-13} m. An independent integration with MATLAB `ode45` reproduced both crossings [6].

The two fitted drag coefficients agree to the displayed precision at 0.02007 m^{-1}. The fitted (β_y, β_z) pairs are $(-0.206, -0.463)$ and $(0.085, -0.251)$ s^{-1}, respectively. At any instant, a component of $\boldsymbol{\beta}$ along $\boldsymbol{u}$ contributes no transverse force. As gravity turns the velocity, that component can influence the later flight. Small adjustments in launch velocity allow different force histories to agree at both early frames.

The two flights yield identical centimetre readings but cross at $Q_A = (2.57378, 2.00251)$ m and $Q_B = (2.62831, 1.93396)$ m, 8.76 cm apart. Any point predictor must give them the same $\widehat{Q}$. The triangle inequality then gives

$$\max\{\|\widehat{Q} - Q_A\|, \|\widehat{Q} - Q_B\|\} \;\geq\; \tfrac{1}{2}\|Q_A - Q_B\| \;\approx\; 0.0438 \text{ m}. \tag{7}$$

This pair establishes nonuniqueness even when the displayed centres are fitted exactly. Its 4.38 cm floor is specific to these two flights. For comparison, mean two-frame errors on separate synthetic

entries were 17.0 cm under rounding and 25.5 cm with tracking errors (Table 4). These empirical averages measure a different quantity from worst-case error for the prescribed observations. Extending the fitted β_x values to the permitted endpoints ± 0.8 s^{-1} increases the crossing separation to 13.98 cm and raises the exact-centre floor to 6.99 cm.

4.2 The same centimetre readings can hide a much larger gap

In a measurement system it is the *displayed coordinates* that have to match. We searched the convex hull of the 3,469 stored goal crossings for a widely separated pair. The two selected flights both use model S, with different admissible winds, launch velocities and transverse-force coefficients. Independent adaptive integrations put every early coordinate strictly inside its required centimetre bin, with the closest coordinate 0.043 mm from a bin edge. Both flights reached the open goal before the ground. Their crossing points were $(2.290, 2.329)$ and $(2.906, 1.644)$ m, separated by 0.922 m. By Eq. (7), **any single prediction from these displayed readings must miss at least one of the pair by 0.46 m**. This is an exhibited worst case. The sample search gives no upper bound on the full feasible set or probability that either flight occurs.

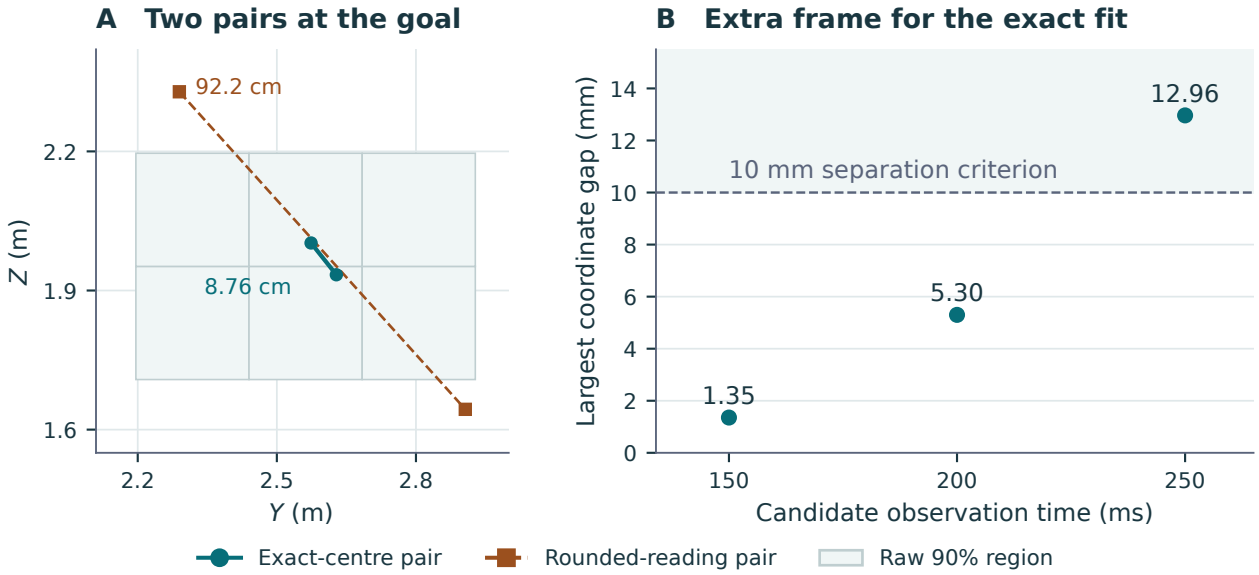


Figure 4. Identical readings can lead to different crossings. A shows one model-S pair fitted to the displayed centres and another that fits within the centimetre rounding intervals. The latter crosses outside the six-cell probability region at both ends. Both axes have the same scale. B follows the original exact-centre pair. With an error bound of 5 mm per coordinate, a third reading separates the pair once its largest coordinate gap exceeds 10 mm.

5 How measurement error affects the forecast

5.1 Position, timing and rounding errors

For axis a, let d_{ia} denote the position in frame i displayed to the nearest $q = 0.01$ m. We model the displayed value as

$$d_{ia} = \mathcal{R}_q[r_a(t_i + \tau; \theta, M) + b_a + e_{ia}], \quad b_a \sim N(0, \sigma_b^2), \quad e_{ia} \sim N(0, \sigma_e^2). \tag{8}$$

Here b_a is a bias common to the frames of one kick, e_{ia} is independent across frames and axes, and τ shifts all the frame times together. Conditional on b, τ, the probability of the displayed readings is

$$L = \prod_{i,a} \left[\Phi\left(\frac{d_{ia} + q/2 - r_a(t_i + \tau) - b_a}{\sigma_e} \right) - \Phi\left(\frac{d_{ia} - q/2 - r_a(t_i + \tau) - b_a}{\sigma_e} \right) \right]. \tag{9}$$

Here Φ is the standard normal cumulative distribution function. With rounding alone, L is the indicator that every coordinate lies in its interval. We sample b from its prior and integrate a shared uniform timing offset by Gauss–Legendre quadrature. The synthetic tests use five or seven nodes. We check the prescribed-kick timing calculation separately below.

A position error small enough to miss by eye can dominate the inferred acceleration. For an exact release coordinate and frame errors on one axis with common variance σ^2 and correlation ρ_e, the acceleration error inferred from the second difference at spacing h has variance

$$\mathrm{Var}\left(\frac{e_2 - 2e_1}{h^2} \right) = \frac{(5 - 4\rho_e)\sigma^2}{h^4}. \tag{10}$$

A common timing shift contributes $-v\tau/h^2$ to the acceleration inferred from uniform motion. At $v = 30$ m/s and $h = 50$ ms, a 1 ms shift produces an apparent acceleration of 12 m/s^2. A spatial bias cancels from the second difference if it affects release and both frames equally. If the release coordinate is exact, a bias shared only by the two frames survives and can resemble aerodynamic curvature.

For the prescribed readings, we allowed a shared phase τ uniform on $[-1, +1]$ ms and refitted launch velocity at each possible pair of true frame times. Weighting the phases by the evidence from both readings gave a median release-to-crossing time of 394 ms and a central 90% interval of 383–406 ms on direct entries. Measured from the *actual second frame*, the corresponding median was 294 ms and the interval was 282–306 ms. The calculation used 24 quadrature nodes and 524,288 proposals per force model per seed, with pooled ESS above 7,600. At 12 nodes the three quantiles of each distribution changed by less than 1 ms. These intervals are conditional on the assumed timing prior.

5.2 Sensitivity of the prescribed forecast

We compared the nominal case with fourteen one-at-a-time variants and refitted the same readings, using the nominal two seeds and 131,072 proposals per model per seed. Table 3 and Fig. 5 use displacement of the continuous conditional mean to track movement independently of the grid. Region areas use the original grid.

For this kick the largest well-supported change came from the measurement assumptions. Doubling the coordinate bound moved the conditional mean by 14.1 cm and enlarged the six-cell region to 15 cells. With tracking error it moved 23.7 cm, and the region grew to 18 cells. The estimated direct-entry probabilities also fell to 95.6% and 93.4%. A fixed +1 ms shift in both observation times moved the mean by 3.73 cm while retaining six cells. A tighter 2.5 mm bound and a −1 ms shift gave ESS of only 68 and 1. Neither 5 ms shift yielded a positive-weight sample.

Five changes to wind, spin ranges and ball mass moved the mean by 0.25–9.08 cm while retaining six cells. Equal area can hide a different hotspot: halving E's spin range moved the modal centre one 0.244 m cell across the goal. Using S alone enlarged the region to seven cells. The 410–450 g mass limits come from the match-ball specification [8]. Early coordinate error is amplified on extrapolation (Appendix A), while the force laws restrict admissible curvature. The ranking here depends on this kick, the grid and the tested ranges.

Table 3. Nominal forecast and nine selected variants with adequate sampling support. Shift compares the continuous means conditional on entry. Area is the raw 90% conditional region. The final column gives the weighted fraction of trajectories entering directly, before conditioning. Fig. 5 includes all fourteen variants.

Changed assumption	Shift (cm)	Area (m^2)	$\widehat{P}(G)$ (%)
Nominal assumptions	0.00	0.357	100.0
Coordinate half-width: 10 mm	14.11	0.893	95.6
Both observation times: +1 ms	3.73	0.357	100.0
Wind cap: 0 m/s	0.83	0.357	100.0
S: halve β component limits	3.97	0.357	100.0
E: halve ω component limits	9.08	0.357	100.0
E: ball mass 410 g	0.25	0.357	100.0
E: ball mass 450 g	0.45	0.357	100.0
S as the sole force model	0.67	0.417	100.0
6 mm independent + 8 mm shared noise	23.70	1.072	93.4

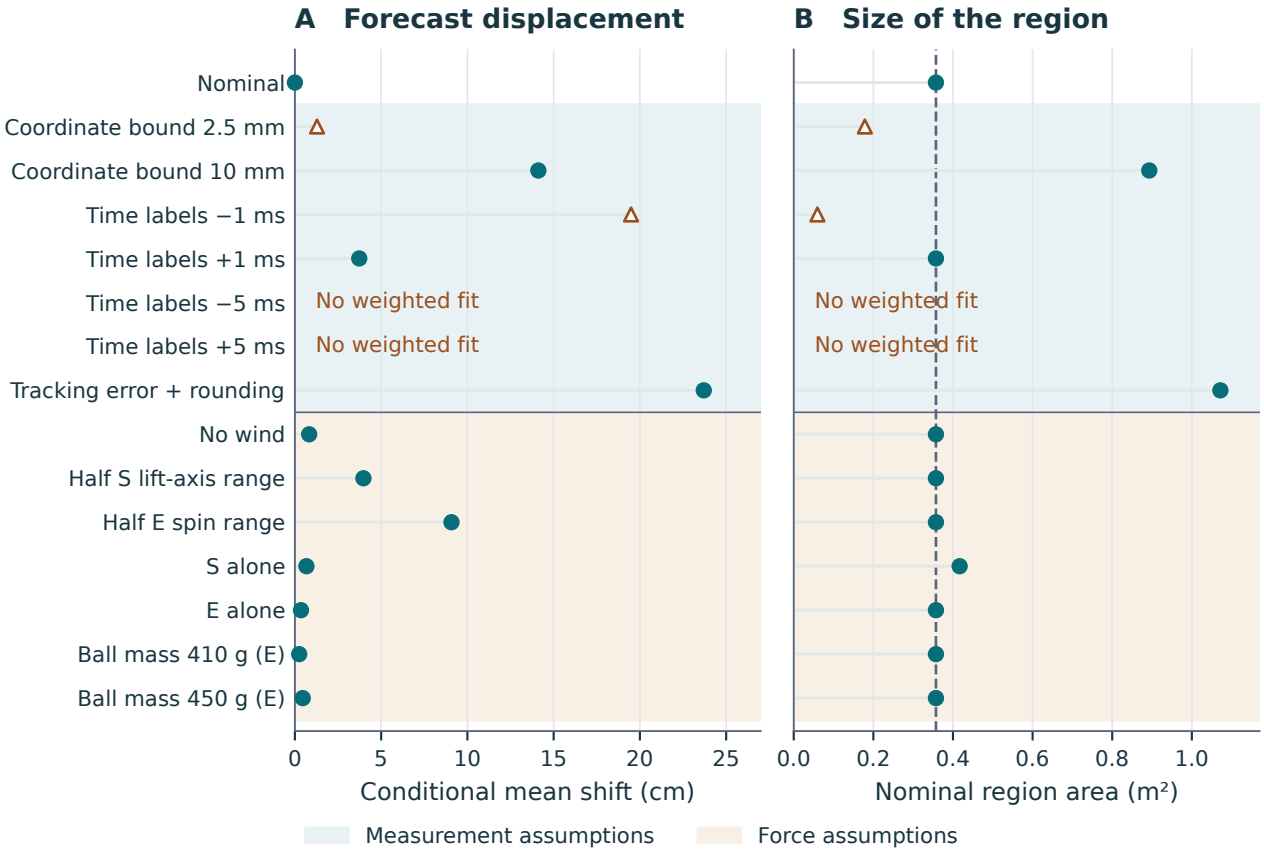


Figure 5. How the expected crossing and region size change. A shows displacement of the conditional mean. B shows the raw 90% region area and its nominal value (dashed). Blue shading groups measurement changes and ochre shading groups force assumptions. Open triangles mark ESS below the prescribed-case threshold of 200. Rows with zero positive weight have no estimate.

6 Testing the two-frame forecast at 100 ms

6.1 Paired forecasts at a common deadline

At 100 ms both aerodynamic predictors know the latest ball position. We give one of them the 50 ms reading as well. Both start computing at 100 ms with the same priors, force laws and solver. Each draws 8,192 proposals per force model, propagates up to 512 positive-weight samples per model and reports the centre of the most probable goal cell. A gravity-only control uses both readings, predicts a continuous ballistic crossing and assigns direct entry a probability of zero or one.

In the first test (Z09) we generated 90 calibration kicks and 120 test kicks with fresh random seeds. The reference solver used a 1 ms step and different force laws from the predictor, with balanced blocks of zero-spin, stable-spin and low-spin wake trajectories (Appendix D). We calibrated the regions on 55 direct entries and scored locations on the 71 test entries. All 120 test kicks contributed to the event scores.

We rounded each kick's exact coordinates to the nearest centimetre. For a second set of readings, we first added independent Gaussian errors in each frame (6 mm standard deviation per coordinate) and a Gaussian bias shared by both frames (8 mm standard deviation per coordinate). Release position and frame times were exact in both sets. For noisy readings the predictor used the same error model as the generator. Appendix A treats timing errors separately.

We prespecified the three methods, coverage targets from 50 to 100% (90% for the main comparison), and subgroups by spin family and distance from the goal edge. Each kick used one inference seed across methods and measurement conditions. The predictor saw only the input positions. We saved and hashed its forecasts before scoring them against the references.

6.2 Calibrating regions on the goal grid

For each method and measurement condition, calibration sets a threshold for including cells. In the aerodynamic forecasts, rank the grid cells $j_1, \dots, j_J$ by decreasing probability $H_D(j)$ conditional on direct entry, breaking ties in a fixed order. The score of an occupied cell is the mass of the cells ahead of it,

$$S_D(j_k) = \sum_{\ell<k} H_D(j_\ell). \tag{11}$$

An empty cell scores one. For gravity, a cell whose centre lies distance d_j from the ballistic crossing scores $d_j/(L + d_j)$, where L is the goal diagonal. Thresholding this score amounts to choosing a radius about the crossing. If the map or ballistic crossing is unavailable, every cell receives a score of one.

For a target coverage c and n calibration entries, sort the scores of their true cells and take the $\lceil (n+1)c \rceil$-th smallest as q_c. Include all cells with $S_D(j) \le q_c$. If the rank exceeds n, set $q_c = 1$. Whenever $q_c = 1$, the region covers the whole goal. This split-calibration rule gives marginal coverage under exchangeability [9]. Our test measures the hit rate on the specified, block-balanced synthetic population. Coverage on real kicks requires independent field data.

6.3 The early frame narrows the region under centimetre rounding

At the 90% target, two frames covered 67/71 entries in 0.381 m^2 on average, compared with 64/71 in 2.003 m^2 using the late frame alone (Table 4). The respective 95% Wilson intervals for coverage were 86.4–97.8% and 81.0–95.1%. At their 95% targets, both methods covered 70/71 entries. The two-frame region averaged 0.585 m^2, compared with 3.482 m^2 for the late frame. Gravity covered 70/71 in 1.222 m^2 at its 90% target (Fig. 6). Equal counts can include different kicks.

Table 4. Forecasts at the 90% calibration target. Hits, mean area and mean point error use the 71 direct entries. The Brier score for direct entry uses all 120 test kicks. The gravity control uses both readings and a calibrated radius.

Profile	Forecast	Hits	Area (m^2)	Mean error (m)	Brier
rounded	100 ms only	64/71	2.003	0.208	0.070
rounded	50 + 100 ms	67/71	0.381	0.170	0.018
rounded	Gravity + radius	70/71	1.222	0.186	0.083
tracking	100 ms only	62/71	1.825	0.216	0.075
tracking	50 + 100 ms	60/71	0.859	0.255	0.078
tracking	Gravity + radius	69/71	1.255	0.199	0.075

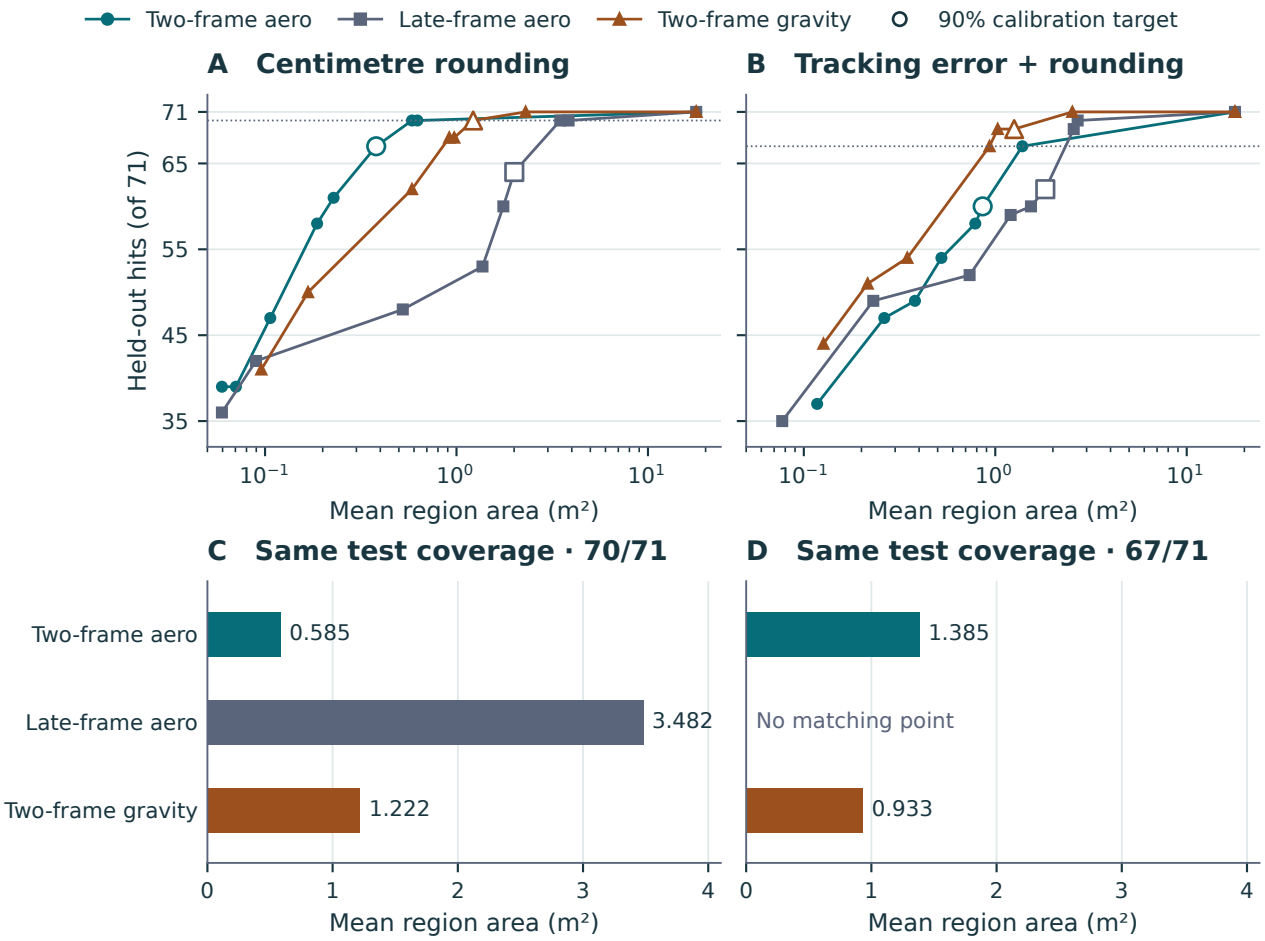


Figure 6. Goal area at matched test coverage. Panels A and B plot hits against mean area on a log scale for the same 71 direct entries. Open markers denote the 90% calibration target. Panels C and D compare prespecified thresholds that yielded equal hit counts. The late-frame curve has no sampled point at 67/71 hits. The full opening spans 17.86 m^2.

Adding the early frame lowered point error on 13 entries, left it unchanged on 54 and raised it on four (Fig. 7). The mean reduction was 3.80 cm (95% interval 0.22–7.62 cm from 4,000 paired resamples of kicks). The mean reduction relative to gravity had a 95% interval of −2.29 to 5.85 cm. The Brier score for direct entry was 0.018 with both frames, 0.070 with the late frame and 0.083 for the binary gravity forecast.

6.4 Tracking error changes which forecast is more precise

At the same 90% target, the two-frame region was the smallest but covered only 60/71 noisy crossings, compared with 62/71 for the late frame and 69/71 for gravity. All eleven misses by the two-frame forecast came from the stable-spin group (12/23 hits). It covered 24/24 in each of the zero-spin and wake groups. Gravity covered 21/23 stable-spin entries (Table 7).

At 67/71 observed hits, gravity used 0.933 m^2 at its 80% target and two-frame aerodynamics used 1.385 m^2 at its 95% target. These two targets yielded equal coverage on this particular test sample. For the 71 direct entries, the aerodynamic grid centre had a mean point error 5.57 cm larger than the continuous gravity forecast (paired 95% interval 1.17–10.25 cm). The intervals for both the point-error and Brier-score differences between two-frame and late-frame aerodynamics included zero.

Estimating acceleration from the readings divides the second position difference by h^2, with $h = 50$ ms. A bias shared by the two frames survives in that difference when the release position is fixed independently (Eq. (10)). Gravity fixes the acceleration. Its relative precision here is consistent with the difficulty of estimating transverse curvature from noisy readings, even when the noise model is known.

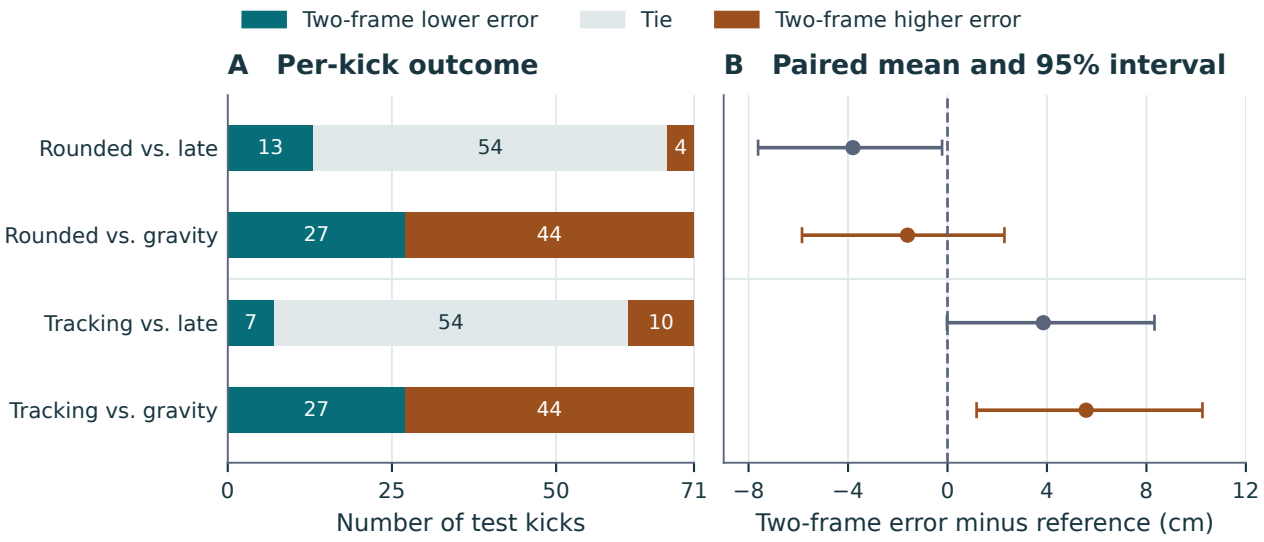


Figure 7. Point errors on the same 71 direct-entry kicks. A counts kicks on which two-frame aerodynamics has a smaller, equal or larger error than each comparison method. B shows the mean paired difference and its 95% shot-bootstrap interval. Negative values favour two-frame aerodynamics.

6.5 Forecast availability and computation time

Each method produced a point for all 71 direct entries and an event probability for all 120 kicks under both measurement conditions. Only one run, a noisy late-frame forecast, fell below 25 effective samples. Even with a defined point, the noisy two-frame region filled the entire goal at its 97.5%

target. Appendix B gives the counts near the goal edge.

In the local Python run, two-frame inference and goal-map construction took 84 ms at the median and 94 ms at the 95th percentile with rounding alone. With tracking errors the times were 121 and 131 ms. At the test sampling budget, 131 ms is about 44% of the nominal 299 ms remaining after the last reading. Capture, reconstruction, transfer and goalkeeper movement require additional time, and the crossing time varies by kick.

6.6 Stable spin in an independent test

The 12/23 aerodynamic hits on noisy stable-spin entries in Z09 prompted a second test (Z10). We drew 300 new calibration kicks and 600 new test kicks, balanced by the same three spin families, and used DOP853 to generate the reference positions and first events. Each predictor saw only the 50 and 100 ms readings. We locked the forecasts, fitted a 90% threshold for each method on the 180 calibration entries pooled across families (rank 163), then scored the held-out crossings.

With tracking errors, two-frame aerodynamics covered 343/368 direct entries overall (93.2%) and 90/110 with stable spin (81.8%). The stable-spin group accounted for 20 of the 25 misses (Fig. 8A). Gravity covered 77/110 stable-spin entries. Aerodynamics alone covered 31 and gravity alone covered 18, a paired difference of +11.8 percentage points (95% interval 0.0 to +23.6). The mean areas were 1.019 and 0.845 m^2, respectively. In a *post hoc* check, we exchanged the previously fitted Z09 and Z10 thresholds. With both forecasts fixed for each of these 110 kicks, the Z09 thresholds gave 78 aerodynamic and 94 gravity hits. Changing only to the Z10 thresholds reversed the order to 90 and 77. On the original 23 entries gravity led under either pair of thresholds (Appendix H).

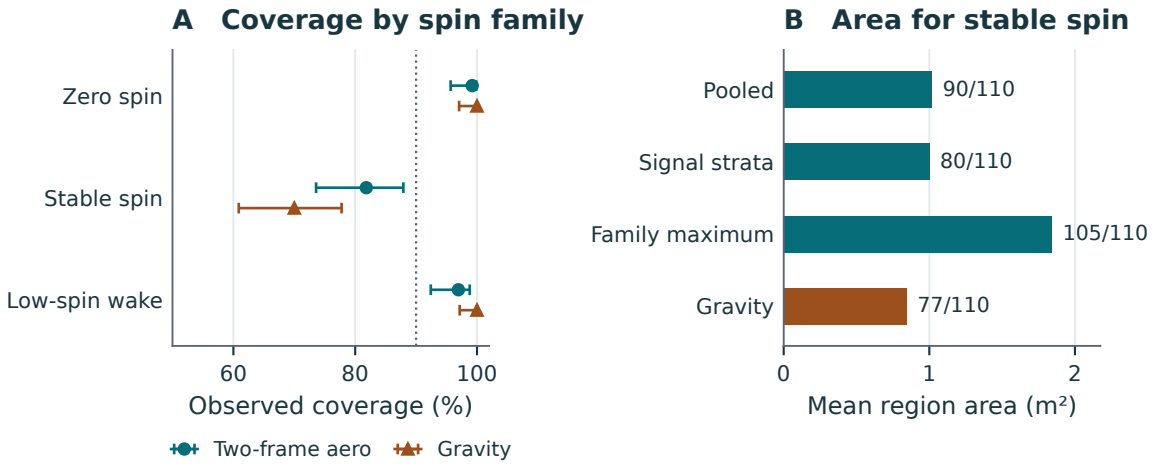


Figure 8. Two-frame regions under tracking noise in Z10. A shows coverage and 95% Wilson intervals for 127 zero-spin, 110 stable-spin and 131 low-spin wake entries. Each method uses its own pooled cutoff. The dotted line marks the 90% calibration target. B gives mean areas and hit counts on the 110 stable-spin entries. The first three bars apply a pooled cutoff, cutoffs by measured curvature, or the largest family cutoff to the same aerodynamic maps. Gravity uses its own pooled radius.

Cleaner measurements improved the point forecast. On these 110 kicks, centimetre rounding alone, with a matching likelihood and calibration, reduced mean aerodynamic point error from 0.421 to 0.258 m and raised coverage to 100/110. Keeping the noisy observations and halving the prior ranges for S's transverse-force coefficient and E's spin vector gave a mean error of 0.379 m, with 94/110 hits in 0.579 m^2 after calibration. We could also expand the original noisy regions by applying the largest of the three family cutoffs to every kick. This rule needs spin labels only in calibration. It recovered

15 stable-spin entries (105/110) at a mean area of 1.843 m^2. Adjusting the cutoff changes which cells enter the region. The point prediction and event probability still come from the original forecast.

7 From a crossing forecast to a decision time

With an exact frame clock the prescribed kick leaves a median of 299 ms for computation and movement after the last reading. Under the shared ±1 ms phase prior, the corresponding estimate is 294 ms (central 90%, 282–306 ms). In a separate synthetic test, shifting the two observations from 50/100 to 100/150 ms reduced mean point error from 0.182 to 0.126 m and left 50 ms less time to act. Keeping all three frames at the later deadline gave more low-ESS forecasts (Appendix E). To translate either forecast into a save probability, one would also need the goalkeeper's initial position, movement onset and reach.

Match tracking poses a different obstacle. Three penalties recorded at 25 Hz had later crossing references, but none had an independently verified release time. The nominal force and measurement assumptions could not explain both early frames in any case. At Regensburg the single +80 ms aerodynamic forecast missed the later crossing reference by 0.849 m, compared with 0.809 m for gravity on the same input. The two-frame record failed the acceleration check. The single-frame comparison cannot estimate field accuracy or calibrate the synthetic two-frame errors (Appendix F).

The forecast describes an uninterrupted flight under specified aerodynamic priors. Ground rebounds, interventions and collisions with the posts or crossbar could change the outcome near the goal edges, and the mix of forces in actual matches may differ from the synthetic population. The grid, measurement model and decision time therefore belong alongside any quoted point and region.

Conclusion

For the prescribed kick, the measured heading turns toward the centre relative to its first 50 ms chord. Under the stated wind and rounding limits the turn requires a transverse aerodynamic force. The modal cell is centred at $(2.56, 2.07)$ m, and six cells cover 0.357 m^2 and contain 90.5% of the conditional entry probability. Two force histories that generate the same displayed readings cross 0.92 m apart, illustrating how a concentrated probability map can coexist with a large worst-case error.

For this kick, less precise coordinates enlarged the region more than the tested changes in wind, lift ranges and mass. Independent synthetic kicks show the consequence for prediction: with tracking errors, the stable-spin group fell below the pooled coverage target. More precise readings also improved the estimated point, while a higher cutoff covered more crossings by admitting a larger part of the goal. Real-penalty validation requires a release time and crossing measured independently of the two early positions.

Appendix A How timing and coordinate errors move the crossing

A.1 From a position error to a goal-plane error

A position error can change both where and when the ball crosses the goal plane. We can separate these effects in the constant-acceleration fit. For a fixed release position and readings at h and $2h$, its acceleration and launch velocity are

$$\widehat{\boldsymbol{a}} = \frac{\boldsymbol{r}_2 - 2\boldsymbol{r}_1 + \boldsymbol{r}_0}{h^2}, \qquad \widehat{\boldsymbol{v}}_0 = \frac{\boldsymbol{r}_1 - \boldsymbol{r}_0}{h} - \frac{h}{2}\widehat{\boldsymbol{a}}. \tag{12}$$

The prescribed readings give $\boldsymbol{v}_0 = (30.4, 9.4, 6.9)$ m/s and $\boldsymbol{a} = (-16, -16, -12)$ m/s^2. Extrapolate with this acceleration and write $X = 11 - x_0$, $D = v_{0x}^2 + 2a_xX$. For a first forward crossing with $D > 0$, the positive root has the numerically stable form

$$T = \frac{2X}{v_{0x} + \sqrt{D}}, \quad \dot{x}(T) = \sqrt{D}. \tag{13}$$

The control crosses at $T = 0.4050$ s and $(Y, Z) = (2.495, 1.920)$ m. The aerodynamic forecast appears in Eq. (6).

Let $\delta\boldsymbol{r}(T)$ be the change in position at the original crossing time. Requiring the perturbed trajectory to satisfy $x = 11$ m at its own crossing gives

$$\delta T = -\frac{\delta x(T)}{\dot{x}(T)}, \qquad \delta Q = P\,\delta\boldsymbol{r}(T), \qquad P = \begin{pmatrix} -\dot{y}(T)/\dot{x}(T) & 1 & 0 \\ -\dot{z}(T)/\dot{x}(T) & 0 & 1 \end{pmatrix}. \tag{14}$$

The local slopes convert longitudinal error into a shift across the goal. An error parallel to the velocity changes only the arrival time to first order, since $P\boldsymbol{v}(T) = 0$. The slopes can diverge as $\dot{x}(T)$ approaches zero. If the forward crossing disappears, its location derivative is undefined.

The Lagrange weights of the two readings are $\lambda_1 = s(2 - s)$ and $\lambda_2 = s(s - 1)/2$, where $s = T/h$. Let $\boldsymbol{e}_i$ be the spatial error in reading i. A common time offset $\tau > 0$ means that we fit the readings at their labelled times $h, 2h$ although both were taken τ later. To first order, the combined shift is

$$\delta Q = P\left[\lambda_1\boldsymbol{e}_1 + \lambda_2\boldsymbol{e}_2 + \big(\lambda_1\boldsymbol{v}(h) + \lambda_2\boldsymbol{v}(2h)\big)\tau\right]. \tag{15}$$

At the nominal crossing the weights are -49.41 and 28.76. Allowing independent errors of up to 5 mm in each coordinate gives linear bounds of 0.439 m in Y and 0.424 m in Z. A common $+1$ ms offset shifts (Y, Z) by $(-0.123, -0.093)$ m to first order. Central differences with a $1\,\mu$s step reproduce the timing derivative to within 2×10^{-5} m/s. The large weights arise from extending an unconstrained quadratic well beyond the observations. The aerodynamic predictor restricts curvature through its force and launch bounds.

With the same positions reassigned to times $h+\tau, 2h+\tau$ and the quadratic refitted, $dT/d\tau = -26.52$ at zero. The exact forward roots for $\tau = -1, 0, +1$ ms are respectively 439.2, 405.0 and 383.1 ms. Holding the time labels fixed and instead generating later positions, as in Eq. (15), reverses the local derivative. This timing sensitivity belongs to the unconstrained quadratic. The aerodynamic timing interval in Sec. 5 weights physically admissible flights.

A.2 Covariance near a smooth crossing

Let Σ_e be the covariance of noise drawn independently in each frame, Σ_b that of a bias shared by the frames, and σ_τ^2 the timing variance. Assume independence among these sources. With $w = \lambda_1 \boldsymbol{v}(h) + \lambda_2 \boldsymbol{v}(2h)$, Eq. (15) gives

$$\mathrm{Cov}(\delta Q) = P\left[(\lambda_1^2 + \lambda_2^2)\Sigma_e + (\lambda_1 + \lambda_2)^2 \Sigma_b + \sigma_\tau^2 w w^\top\right] P^\top. \tag{16}$$

Timing uncertainty acts along a single direction, Pw. The bias coefficient changes if the release shares the frames' translation. This covariance describes measurement uncertainty near a smooth crossing. Model discrepancy and sampling error contribute separately.

A.3 Joint perturbations and loss of a forward crossing

We combined both timing limits and zero offset with every corner of the six-coordinate error box, counting a zero bound once. For each combination we refitted the quadratic and solved for its first forward plane crossing, ignoring the ground and goal edges. The largest displacement among sampled crossings need not bound the full box, whose interior may contain larger shifts.

Table 5. Joint perturbations of the quadratic control. The two linear bounds sum the absolute derivative contributions in each coordinate. The sampled shift is the largest Euclidean displacement among forward crossings. The last column counts trials with no forward root among all tested combinations.

Phase bound (ms)	Spatial bound (mm)	First-order Y (m)	First-order Z (m)	Sampled shift (m)	No root (count)
0	0	0.000	0.000	0.000	0/1
0	2.5	0.219	0.212	0.319	0/64
0	5	0.439	0.424	0.671	0/64
0	10	0.877	0.848	1.502	0/64
1	0	0.123	0.093	0.197	0/3
1	2.5	0.342	0.305	0.533	0/192
1	5	0.562	0.517	0.930	0/192
1	10	1.000	0.942	2.025	0/192
3	0	0.370	0.280	0.297	1/3
3	2.5	0.589	0.492	0.537	64/192
3	5	0.808	0.704	2.298	48/192
3	10	1.247	1.128	4.487	32/192

With bounds of 1 ms and 5 mm, all 192 combinations crossed the plane. The largest sampled displacement was 0.930 m, while the separate linear bounds in Y and Z were 0.562 and 0.517 m (Table 5). At 3 ms and 5 mm, 48/192 combinations lost the forward root. Widening a box changes the corners sampled, so the failure fraction can rise or fall. Propagating each perturbation to its own crossing captures both large shifts and the loss of the event.

Appendix B Coverage by subgroup and computation time

Mean region area includes all 71 direct entries among the 120 test kicks. A calibration target of 100% returns the full opening by construction.

We classify a crossing as near-edge if its distance to the nearest of the four goal edges is below one nominal cell width (0.244 m). The generator family labels are used only to score subgroups. Each kick contributed one inference seed per measurement condition. We matched differences by kick and

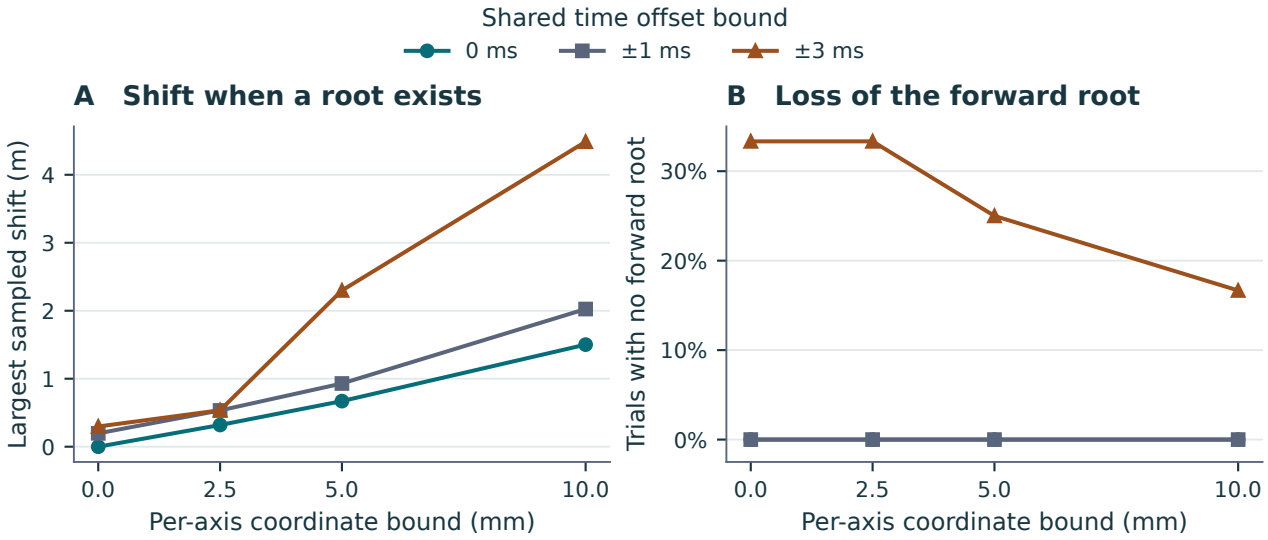


Figure 9. A perturbation can shift or remove the plane crossing. A gives the largest displacement among sampled quadratic fits with a forward root. B gives the fraction of tested combinations with no root. Each timing and coordinate combination is a deterministic trial.

Table 6. Observed hits and mean area at selected calibration targets. Both measurement conditions use the same 71 direct entries.

Profile	Forecast	Target (%)	Hits	Mean area (m^2)
rounded	100 ms only	80	53/71	1.367
rounded	100 ms only	90	64/71	2.003
rounded	100 ms only	95	70/71	3.482
rounded	100 ms only	100	71/71	17.861
rounded	50 + 100 ms	80	58/71	0.187
rounded	50 + 100 ms	90	67/71	0.381
rounded	50 + 100 ms	95	70/71	0.585
rounded	50 + 100 ms	100	71/71	17.861
rounded	Gravity + radius	80	68/71	0.917
rounded	Gravity + radius	90	70/71	1.222
rounded	Gravity + radius	95	71/71	2.296
rounded	Gravity + radius	100	71/71	17.861
tracking	100 ms only	80	59/71	1.203
tracking	100 ms only	90	62/71	1.825
tracking	100 ms only	95	69/71	2.569
tracking	100 ms only	100	71/71	17.861
tracking	50 + 100 ms	80	54/71	0.523
tracking	50 + 100 ms	90	60/71	0.859
tracking	50 + 100 ms	95	67/71	1.385
tracking	50 + 100 ms	100	71/71	17.861
tracking	Gravity + radius	80	67/71	0.933
tracking	Gravity + radius	90	69/71	1.255
tracking	Gravity + radius	95	71/71	2.527
tracking	Gravity + radius	100	71/71	17.861

Table 7. Hits at the 90% target, by generator family and distance to the edge of the opening. Each grouping partitions the 71 entries. All subgroups use the threshold fitted for that method and measurement condition.

Profile	Stratum	Entries	100 ms	50 + 100 ms	Gravity
rounded	no spin	24	24/24	24/24	24/24
rounded	stable spin	23	16/23	21/23	22/23
rounded	wake	24	24/24	22/24	24/24
rounded	near edge	11	9/11	10/11	11/11
rounded	interior	60	55/60	57/60	59/60
tracking	no spin	24	24/24	24/24	24/24
tracking	stable spin	23	14/23	12/23	21/23
tracking	wake	24	24/24	24/24	24/24
tracking	near edge	11	9/11	7/11	10/11
tracking	interior	60	53/60	53/60	59/60

resampled 4,000 times, using all kicks for event scores and direct entries for point error, area and hits. The intervals describe the sampled mixture of kick families, balanced before outcomes were known. Field calibration would require independently timed releases, reference crossings and a protocol for intervening contacts.

Table 8. Availability and elapsed time in one local run. Aerodynamic timings cover inference and preparation of the goal-cell masses for ranking. Gravity timings cover the ballistic solution. Both exclude application of the calibrated threshold.

Profile	Forecast	Points	Events	ESS < 25	Median (ms)	95th (ms)
rounded	100 ms only	71/71	120/120	0	90.9	103.1
rounded	50 + 100 ms	71/71	120/120	0	83.6	93.7
rounded	Gravity + radius	71/71	120/120	0	0.2	0.3
tracking	100 ms only	71/71	120/120	1	107.5	117.4
tracking	50 + 100 ms	71/71	120/120	0	121.4	130.7
tracking	Gravity + radius	71/71	120/120	0	0.2	0.3

Appendix C Forecast algorithm and physical checks

C.1 From observations to first-event probabilities

The inputs are the release point, observed positions and times, the camera-error model and the parameter ranges. Each observation must have a valid time index and finite coordinates.

1. Draw wind and aerodynamic parameters for each force law. Propose launch velocities by inverting the latest observed position, mixing in 2% draws from the launch prior to retain support elsewhere. With only the release point, sample directly from that prior.

2. Propagate each trajectory through the available observations. A trajectory that reaches the goal plane or touches down before the last reading receives zero weight. Use the observation likelihood at all supplied times in Eq. (5). If release timing is uncertain, divide the fixed budget among quadrature nodes and combine posterior masses with their quadrature weights.

3. Propagate the surviving trajectories to the first forward crossing, descending ground contact or time horizon. Normalize each event mass by the total weight. A crossing in the open goal rectangle contributes to its goal cell. Divide each cell mass by the total direct-entry weight.

4. Return the event probabilities, modal cell, raw region and effective sample size. Zero total weight leaves the event distribution undefined. If only the direct-entry weight is zero, the event probabilities remain available and the conditional point and map are undefined.

Report the unconditional entry probability alongside the conditional point, especially near the goal boundary where entry mass may be small. Geometric bounds can exclude a flight under the stated assumptions. A low effective sample size flags uncertainty in the numerical approximation.

C.2 A bound on the force needed to turn the horizontal flight

Put $L = 2h$, $\boldsymbol{a}_* = (\boldsymbol{p}_2 - 2\boldsymbol{p}_1)/h^2$ and let $\boldsymbol{n}$ be the left unit normal to the chord $\boldsymbol{p}_2 - L\boldsymbol{w}$ in the air frame. Integrating by parts gives $\boldsymbol{n}\cdot\boldsymbol{a}_* = \int_0^L \min(t, L-t)\,\boldsymbol{n}\cdot\ddot{\boldsymbol{p}}(t)\,dt/h^2$. The second difference therefore measures a time-weighted net acceleration. To minimize its inward magnitude over a wind disk of radius W, rotate the chord by the largest possible angle to the left. Its heading is then $\arg(\boldsymbol{p}_2) + \arcsin(LW/\|\boldsymbol{p}_2\|)$, with $\boldsymbol{n}\cdot\boldsymbol{p}_2 = -LW$. For these positive coordinates and negative acceleration components the inward projection decreases as this angle increases. Over the rounding box it is smallest when x_1 is largest, y_1 smallest, x_2 smallest and y_2 largest. The limiting points in the closed box are $(1.505, 0.445)$ and $(2.955, 0.865)$ m.

To separate the non-drag force from the net acceleration, write $d(t) = \boldsymbol{n}\cdot[\boldsymbol{p}(t) - t\boldsymbol{w}]$, $A = -\boldsymbol{n}\cdot\boldsymbol{a}_* = 2d(h)/h^2$ and $b(t) = At(L-t)/2$. The horizontal equation is $d'' + c(t)d' = \ell_n(t)$, with c the drag rate and ℓ_n the projected non-drag acceleration. The difference $d - b$ vanishes at $0, h, L$. Rolle's theorem applied first to $d - b$, then to $e^{\int_0^t c}(d-b)'$, gives a time s such that $-\ell_n(s) = A[1 - c(s)(h-s)]$. Over the first 100 ms, $\dot{U} \le g$ gives $U \le \|(38, 12, 11)\| + 3 + 0.1g < 45.33$ m/s. With the coefficients from Sec. 2.1, $c \le \max(\kappa_{\max}, KC_{D,\max})U < 1.854$ s^{-1}. The peak inward non-drag component must therefore reach at least $(1 - 1.854h)A$, about 8.09 m/s^2 at the displayed centres and 0.97 m/s^2 allowing rounding.

C.3 Assumptions, limits and solver checks

Table 9. Inputs used in the nominal forecast and in the tracking-error test. A change to an assumed range or error model requires a new forecast.

Quantity	Role	Interpretation
Release, frame times and opening	Given by the problem	Fix the observations and geometry
Ball mass, radius and air density	Nominal physical inputs	Set the factor $\rho\pi R^2/(2m)$
Empirical drag curve	Literature-based fit	Limited speed range, with clipping and an offset prior
Wind and lift or spin bounds	Scenario choices	Restrict admissible flights and affect model weights
5 mm coordinate half-width	Rounding assumption	Links continuous positions to displayed readings
Cell width and RK4 steps	Numerical choices	Affect the modal cell and integration accuracy
6 mm noise and 8 mm bias	Synthetic test settings	Specify independent and shared tracking errors

Under gravity alone, $x(t) = v_{0x}t$, $y(t) = v_{0y}t$ and $z(t) = R + v_{0z}t - gt^2/2$. The analytic plane and ground times check event ordering. In still air, drag alone preserves horizontal heading because the horizontal acceleration is proportional to the horizontal velocity. Equation (2) extends this property to the frame of a constant wind. These limits check the geometry and force signs independently of posterior sampling.

Four RK4 comparisons with adaptive DOP853 cover vacuum flight, drag with ground contact, effective transverse lift and the empirical force law. The solvers agreed to within 0.171 mm in event position and 0.038 ms in event time on these cases. The original exact-centre pair in Sec. 4 also has a MATLAB replay. Independent adaptive integrations verify the two rounded-coordinate witnesses. The observations still limit what can be inferred about wind, spin and release time.

Appendix D Reference flights and earlier synthetic tests

D.1 The reference generator and kinematic controls

To test how the forecast responds to imperfect force laws, we generated reference flights with a different drag curve,

$$C_D^{\mathrm{ref}} = 0.19 + \frac{0.20}{1 + \exp[(U - 20)/2.5]} + d_0, \qquad d_0 \sim \mathcal{U}(0, 0.1),$$

and a lift coefficient $0.32 \tanh(4R\|\boldsymbol{\omega}\times\boldsymbol{u}\|/U^2)e^{-0.12t}$ acting along $\boldsymbol{\omega}\times\boldsymbol{u}$. Launch-velocity components were independently uniform on $[20, 36]$, $[-11, 11]$, $[2, 10]$ m/s. Wind was uniform on a horizontal disk of radius 2.5 m/s. We used equal numbers of kicks with zero spin, stable spin and a low-spin wake. Their spin-component bounds were 0, 55 and 6 rad/s. Wake flights also had a time-varying acceleration given by projecting $A(0, \sin\phi(t), 0.4\cos\phi(t))$ perpendicular to $\boldsymbol{u}$. We chose $A \in [2, 5]$ m/s^2, a period of 0.18–0.55 s and a uniform initial phase. Experiments motivate including unsteady low-spin forces [7]. The amplitudes, periods and family proportions here are synthetic.

The original study used DOP853 [5] and split whole kicks into 9 for development, 36 for calibration and 48 for blind testing. Follow-up tests through Z09 used new seeds with the same reference force laws and a 1 ms RK4 step. Z10 used DOP853 for its new reference flights. On the original 48

DOP853 flights, RK4 reproduced every first event, with maximum differences of 1.9×10^{-6} m at the crossing and 6.3×10^{-7} s in event time. We gave each predictor only its designated early frames and saved and hashed its forecasts before scoring. These comparisons test numerical agreement and performance on the simulated kick population.

On true direct entries we report Euclidean point error, region hits and mean region area. We score the probability of direct entry across first events with the Brier score $(P(G) - \mathbf{1}_G)^2$. We compare against a least-squares fit under gravity alone and a constant-acceleration fit through the three positions. We count missing event forecasts separately from missing conditional points.

D.2 The early frame at a fixed sampling budget

Of the original 48 blind-test kicks, 33 entered directly, ten crossed outside the opening and five hit the ground first. The two-frame method had median point error 0.133 m and raw-region coverage 30/33, using eight times the single-frame proposal budget. We revisited the same kicks with 2,048 proposals per model for each input choice and three paired seeds (Table 10).

Table 10. Retrospective test at 100 ms with equal proposal budgets. The 33 direct entries and three seeds give 99 point forecasts. Both methods returned event estimates on all 144 attempts. Regions use raw probabilities.

Observed frames	Mean error (m)	Hits	Area (m^2)	ESS < 25
100 ms only	0.215	99/99	3.071	0
50 and 100 ms	0.170	91/99	0.288	11

Adding the 50 ms reading narrowed the regions and reduced their coverage. We averaged paired differences over seeds within each kick, then resampled the 33 kicks 2,000 times. The mean point-error difference (both frames minus the late frame) was -4.46 cm, with a descriptive 95% bootstrap interval of $[-9.14, 0.08]$ cm. The equal-budget result supports a reduction in area. The interval still admits zero improvement in mean point error.

D.3 When tracking noise exceeds the rounding interval

We applied three likelihoods to identical noisy positions on a separate set of 36 calibration and 48 test kicks. Before centimetre rounding we added an 8 mm Gaussian bias shared across frames for each axis and independent 6 mm Gaussian frame errors. All aerodynamic forecasts used 8,192 proposals per model and two paired seeds. One likelihood treated the combined 10 mm marginal error as independent, one used the simulated 6/8 mm decomposition, and one allowed only rounding. The two tracking models used a local velocity-proposal half-width of $0.026\,\mathrm{m}/t$. The rounding-only model used $0.007\,\mathrm{m}/t$. Importance weights corrected each proposal. The finite-budget comparison therefore includes both the likelihood and the proposal width.

Table 11. Noisy-input tests at 100 ms on 36 direct entries and two seeds. n_Q counts available point forecasts. Region hits and mean area use all 72 attempts. An undefined map gives an empty raw region. A calibrated region can encompass the whole opening.

Likelihood	Mean error (m)	n_Q	Raw hits	Raw area (m^2)	Calibrated hits	Cal. area (m^2)
Rounding only	0.425	68	27/72	0.355	72/72	17.861
Independent error	0.231	72	72/72	1.870	68/72	1.261
Shared + independent	0.257	72	71/72	1.256	71/72	1.609

We calibrated each method and seed separately using the prefix-mass score in Sec. 6. A true cell with zero mass receives score one. With n calibration entries, we include every cell whose score is at most the $\lceil 0.9(n+1) \rceil$th ordered calibration score. A cutoff of one or a rank above n returns the whole opening. Coverage is empirical here because the design balances synthetic families and the field error distribution is unknown.

The rounding-only method covered all 72/72 entries after calibration by returning the entire 17.861 m^2 goal. The shared-bias likelihood reduced mean point error relative to rounding alone by 15.86 cm on 34 complete paired kicks (95% shot-bootstrap interval 4.95–26.53 cm). Its interval against the independent-error model includes zero. Gravity had lower mean point error (0.220 m) than either tracking model, but its event Brier score was higher (0.104 versus 0.069 for independent errors and 0.064 for shared bias). The rounding-only Brier score was 0.101 on 94/96 attempts. The other methods supplied all 96 event estimates.

Appendix E The value and cost of waiting for another frame

E.1 Moving the last frame or keeping an extra one

Using these same calibration and test kicks with rounding alone, we compared three sets of observations at 8,192 proposals per model and two paired seeds. Moving from the 50/100 ms pair to the 100/150 ms pair delays the decision. Comparing that later pair with all three frames holds the 150 ms deadline fixed.

Table 12. Additional observations on 36 direct entries with two seeds (72 attempts). n_Q counts available point forecasts. An undefined map contributes a miss and zero area. Missing event estimates and low ESS use all 96 attempts.

Frames (ms)	Mean error (m)	n_Q	Hits	Area (m^2)	No event estimate	ESS < 25
50, 100	0.182	72	61/72	0.346	0	0
100, 150	0.126	72	62/72	0.136	0	20
50, 100, 150	0.130	65	52/72	0.093	6	46

Waiting for the later pair reduced mean point error by 5.65 cm on 36 complete paired kicks (95% shot-bootstrap interval 2.79–8.98 cm). Mean time remaining fell from 321 to 271 ms. Calibrated coverage rose from 61/72 to 64/72 while mean area fell from 0.350 to 0.177 m^2.

At the same 150 ms deadline, keeping all three frames left six of 96 attempts without an event estimate and 46/96 with low ESS. Both seeds yielded paired points on only 31 kicks. On those kicks the three-frame mean point error exceeded the later pair's by 0.27 cm (95% interval $[-1.71, 2.28]$ cm). The three-frame raw regions were smaller but covered only 52/72 entries. Few proposals fit all three readings. This limits what the comparison can establish about the earliest frame's information content. Model error and finite-test variation may also contribute. An adaptive or sequential sampler would need its own validation.

E.2 When a later frame distinguishes the two fitted flights

At 150, 200 and 250 ms, the largest coordinate gaps in the pair from Sec. 4 are 1.35, 5.30 and 12.96 mm (Fig. 4B). Deterministic centimetre rounding already gives different heights at 200 ms, 1.26 and 1.25 m. If each coordinate can carry any error within ±5 mm, the two reading boxes still overlap at that time. Of the three times tested, only 250 ms distinguishes the flights for every allowed error.

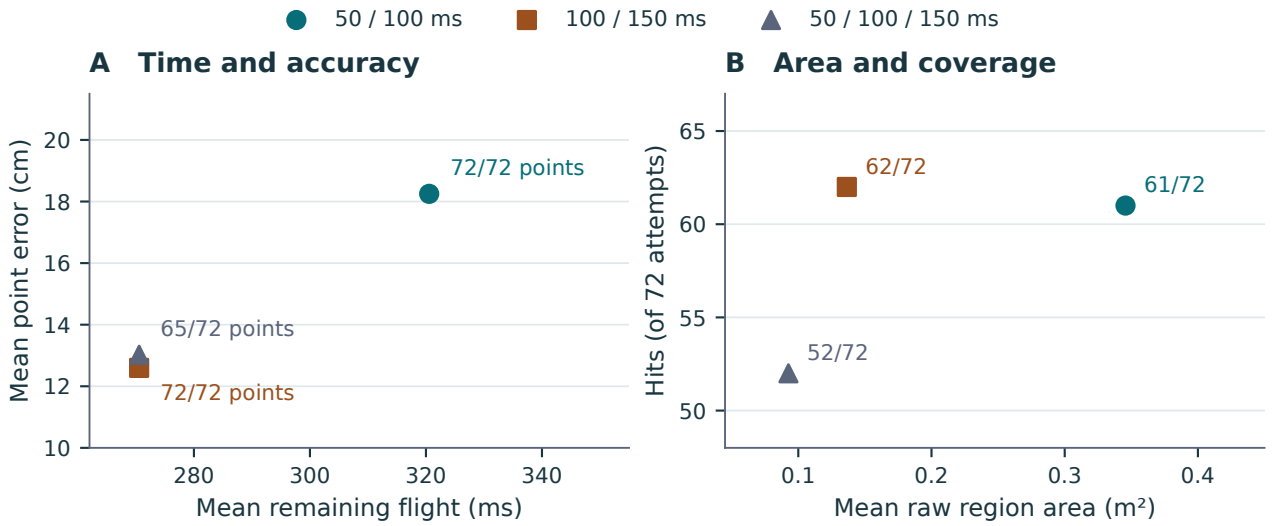


Figure 10. Forecast accuracy and time left to act. A shows mean point error for available direct-entry estimates (65/72 for three frames) and mean remaining flight time over all 36 direct entries, before computation or goalkeeper movement. B shows raw region area and coverage, with undefined maps counted as empty (Table 12).

E.3 Which frames reach the predictor in time?

We simulated 13 camera and noise conditions on a separate set of 48 kicks, 16 per force family and 33 direct entries. At each deadline on the estimated release clock (50 and 100 ms), we selected the latest distinct frame already received and at most 10 ms old. The predictor used its native timestamp. With 2 ms delivery delay, rates of 25, 60, 120 and 240 Hz supplied both readings on 0, 22, 47 and 48 kicks, respectively. At 240 Hz, increasing the delay to 6 ms reduced availability to 46/48. These counts depend on the simulated phase and frame-age rule. Computation takes additional time.

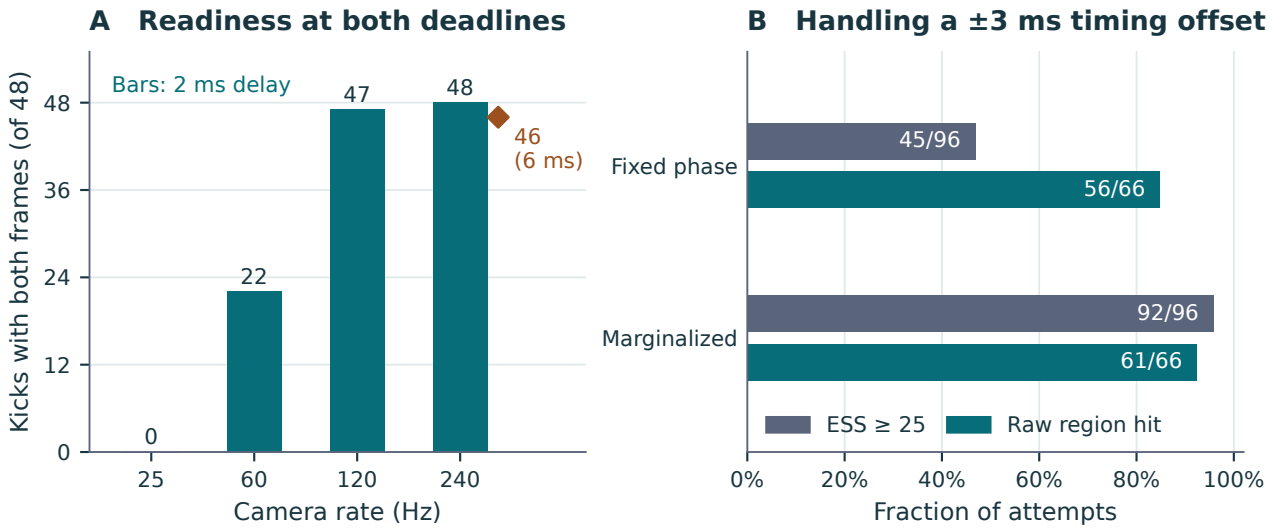


Figure 11. Timely frames and uncertain release times. A counts how many of the 48 kicks supplied both frames on time. B compares a fixed release estimate with integration over the simulated ±3 ms offset. ESS uses 96 attempts (48 kicks and two seeds). Raw-region coverage uses 66 direct-entry attempts.

For periodic frames delivered with fixed delay ℓ and no drops, the oldest possible latest frame has age bounded by $\ell + 1/f$, where f is the frame rate. To guarantee the 10 ms age limit at 2 ms delivery

delay requires at least 125 Hz. A 240 Hz camera allows more margin, but consecutive exposures alone bracket release only to a half-width of about 2.08 ms. A 1 ms timing target would require an independent, finer time reference.

Integrating over the same simulated ±3 ms release offset increased the number of forecasts with ESS at least 25 from 45/96 to 92/96. Raw regions covered 61/66 entries, up from 56/66. They also grew from 0.377 to 0.543 m^2 on average, and mean point error rose from 0.196 to 0.218 m. These simulated conditions provide targets for a field trial.

Appendix F Compatibility with 25 Hz match tracking

F.1 Reference crossings on the match clock

The seven matches in the IDSSE v1 elite-soccer dataset contained three executed penalties, all recorded as goals [10]. We translated the longitudinal coordinate by 41.50 m into the penalty-spot frame. We treated the first live frame after a dead-ball interval as an exact anchor and evaluated forecasts at +80 ms using the original priors and a 5 mm coordinate bound. The inputs were either the +80 ms frame alone or the pair at +40 and +80 ms. Reference crossings were interpolated from later frames in the same system, at least 0.32 s after the cutoff. We locked the forecasts before scoring, but had seen the crossings while screening the data.

Table 13. Three penalties on the match clock. The tracked crossing is the reference. The displacement or second difference in the third column yields the required minimum acceleration under the launch assumptions. Here $\Delta^2 x = x_2 - 2x_1 + x_0$.

Fixture	Reference (Y, Z) (m)	Observed quantity (m)	Min. accel. ($\mathrm{m/s^2}$)
Cologne–Bayern	$(2.750, 0.160)$	$z_1 - z_0 = -0.030$	106.25
Düsseldorf–Regensburg	$(2.919, 1.179)$	$\Delta^2 x = 0.510$	309.38
Düsseldorf–Kaiserslautern	$(3.070, 0.466)$	$\Delta^2 x = 1.100$	678.13

F.2 An acceleration bound for consecutive positions

In constant wind the energy balance gives $\dot{U} \le g$, so $U \le \sqrt{38^2 + 12^2 + 11^2} + 3 + 9.81(0.080) = 45.125$ m/s over the first 80 ms. Drag and lift are orthogonal in Eq. (3). Combining them in quadrature and adding g gives upper acceleration bounds of 58.681, 89.170 and 102.648 $\mathrm{m/s^2}$ for D, S and E, using $C_D \le 0.771$ and $C_L \le 0.38$ in E. Thus $A_* = 102.648$ $\mathrm{m/s^2}$ bounds all three models.

For three coordinates q_0, q_1, q_2 at spacing $h = 0.040$ s, the true second difference integrates acceleration against a triangular kernel of area h^2. An exact anchor and errors bounded by ϵ in the two later positions therefore require

$$|q_2 - 2q_1 + q_0| \le A_* h^2 + 3\epsilon, \qquad z_1 - z_0 \ge v_{z,\min} h - \tfrac{1}{2} A_* h^2 - \epsilon. \tag{17}$$

The second inequality uses the launch prior $v_{z,\min} = 1.5$ m/s. Regensburg's successive x-increments of 0.71 and 1.22 m require at least $(0.510 - 0.015)/h^2 = 309.38$ $\mathrm{m/s^2}$. Kaiserslautern's record repeats the anchor and then advances 1.10 m. Cologne misses the vertical bound by just 2.9 mm. All three exceed the common acceleration cap under the assumed launch conditions (Table 13).

Treating the anchor as a ball already in flight helps distinguish these failures. Relaxing the lower bound on vertical velocity from +1.5 to −11 m/s gives Cologne two forecasts with ESS 58 and 48. The other two records remain unsupported. Cologne is sensitive to the assumed initial velocity, and its release time remains unknown.

For scale, if only the middle frame were mistimed and the longitudinal speed were fixed at 30 m/s, its clock shift δ would produce $\Delta^2 x = -2v_x\delta$. The observed second differences of +0.51 and +1.10 m would correspond to middle frames 8.5 and 18.3 ms early. Moving all three time labels together leaves the spatial second difference unchanged. A repeated position, a mid-flight anchor or a tracking error can produce incompatible readings as well. The records do not identify a clock offset.

F.3 A forecast from the late frame alone

Using only the +80 ms position, neither seed gave positive weight for Cologne or Kaiserslautern. Both placed the Regensburg modal cell at (2.074, 1.098) m, 0.849 m from the reference. The first raw region covered 4.941 m^2 and contained the reference. The second covered 4.584 m^2. With the same late frame, gravity and straight-line fits missed by 0.809 and 1.219 m. With both frames, gravity missed by 0.853 m at Regensburg and predicted ground contact first in the other two cases.

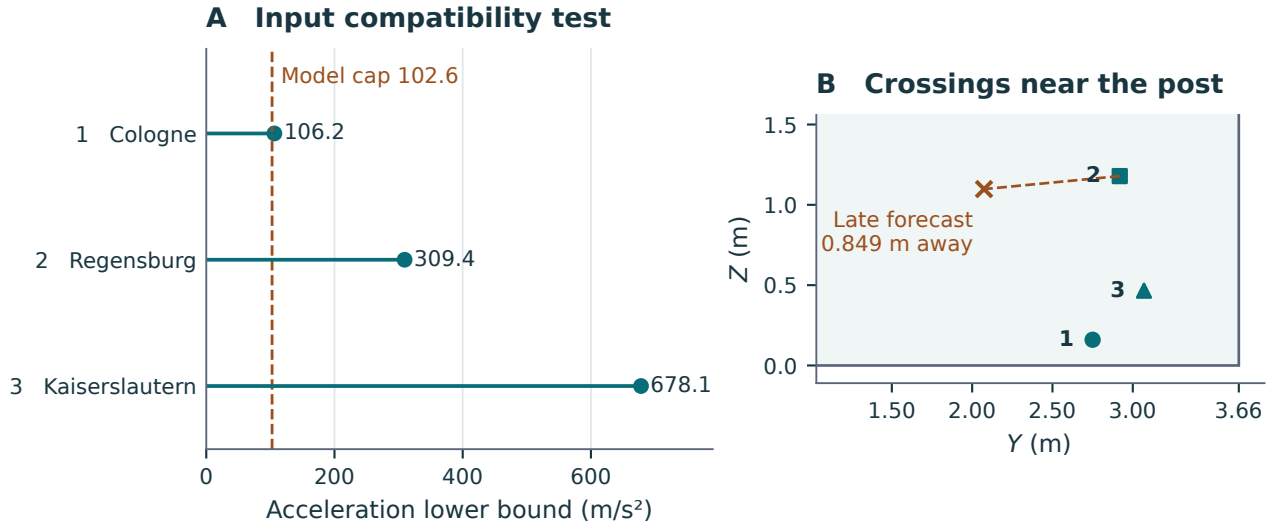


Figure 12. Testing the launch assumptions against three match records. A compares the required minimum accelerations with the common model cap. B locates the three later crossing references near the positive-Y post. The cross marks the only available late-frame aerodynamic forecast. The records leave release timing and the first event unverified.

Appendix G Evidence needed for a field test

Regensburg's anchor lay about 7 cm horizontally from the penalty spot. The other anchors were already 0.71 and 1.47 m downfield. None of the three kicks had an independently timed release or a confirmed first event. ISSIA-3D [11] provides a different check on track quality. Of 79 ordinary-play segments, 47 supplied two positions 80 ms apart and a position 120 ms later. Constant-velocity extrapolation gave a median three-dimensional error of 0.162 m on those 47, compared with 0.996 m for holding the last position.

A field test needs synchronized views of release, calibrated early positions and an independent crossing reference on the same kick. Surface markers can supply spin and deflection together [12]. Record goalkeeper contacts, rebounds, lost frames and the time when each observation becomes available. Estimate timing and spatial errors on separate calibration kicks. On held-out matches, report forecast availability, first-event scores and, among direct entries, point error, coverage and area. Our ball-centre equations omit impacts and deformation. Field tests should also measure wind, ball properties, spin and processing time.

Appendix H Why two-frame forecasts miss some stable-spin entries

In Z10, both the nominal aerodynamic forecast and gravity gave points and regions for all 368 direct entries. With tracking errors, their Brier scores for direct entry were 0.064 and 0.093 over all 600 test kicks. Among the 110 stable-spin entries, the 95% Wilson intervals for coverage were 73.6–87.9% for aerodynamics and 60.9–77.8% for gravity. We calculated paired 95% intervals by resampling whole kicks 4,000 times with the calibration set fixed. The generator balanced the families before flight. Restricting comparisons to direct entries can leave different launch conditions in each family.

H.1 What the two readings tell us about acceleration

For an exact release point and equally spaced observations, write $\widehat{\boldsymbol{a}} = (\boldsymbol{r}_2 - 2\boldsymbol{r}_1 + \boldsymbol{r}_0)/h^2$, with $h = 0.050$ s. Approximating rounding to step q by independent uniform errors gives the per-axis variance

$$\sigma_a^2 = \frac{5(\sigma_e^2 + q^2/12) + \sigma_b^2}{h^4}.$$

The independent errors have squared coefficients $1 + 4 = 5$, and the shared frame bias has coefficient $1 - 2 = -1$. The resulting standard deviation is 2.58 m/s^2 for centimetre rounding alone and 6.76 m/s^2 with the added tracking errors. Across all 200 stable-spin kicks, the observed root-mean-square errors in the second difference, pooled over the y, z components, were 2.56 and 6.47 m/s^2. The aerodynamic point forecast on the 110 direct entries improved by 16.3 cm on average (95% interval 12.7–20.0 cm) when using rounding alone, with a likelihood and proposal width matched to each observation condition.

With release and the noisy 100 ms position fixed, a change $\delta\boldsymbol{a}$ in the second difference changes a constant-acceleration trajectory at a fixed time t by $t(t - 2h)\,\delta\boldsymbol{a}/2$. At 0.40 s, 1 m/s^2 corresponds to 6 cm. We substituted the exact simulated second differences on *all three axes* and solved again for the first plane crossing or ground contact. The forward acceleration can change the crossing time. Of the 110 true entries, 104 had plane crossings in both fits. For these the mean crossing error fell by 45.2 cm (95% interval 38.4–52.8 cm). Five had a crossing only after substitution and one only before. This sensitivity calculation requires exact positions available to the simulator.

H.2 Changing the prior and fitting its threshold

We multiplied the component half-widths of both $\boldsymbol{\beta}$ in S and $\boldsymbol{\omega}$ in E by 0.5 or 1.5. A separate test changed only the D/S/E prior weights from 0.20/0.40/0.40 to 0.02/0.49/0.49. Table 14 shows which effects survive when every prior uses the same threshold. Halving both ranges improved the true cell's mean rank from 11.92 to 6.31 and reduced mean point error by 4.17 cm (95% interval 2.15–6.28 cm). After refitting the cutoff, the corresponding hit-rate difference was +3.6 percentage points (95% interval −3.6 to +10.9). The joint change leaves the individual contributions of S and E unresolved.

H.3 Applying alternative cutoffs to a fixed forecast

The stable-spin calibration entries gave the largest family cutoff (0.9698, rank 51 of 55). A rule that knows each test kick's true family could use its corresponding cutoff. The maximum rule in Fig. 8B applies 0.9698 to every kick. These rules agree on stable-spin entries. Across all 368 entries, using the maximum raised mean area from 1.003 to 1.839 m^2. We also assigned each kick to one of two calibration groups by comparing the measured signal $\|(\widehat{a}_y, \widehat{a}_z + g)\|/\sigma_a$ with a prespecified boundary of 2. This rule uses only the two readings. Relative to pooling, it gained seven stable-spin hits and lost seventeen, covering 80/110 in 1.001 m^2 on average.

Table 14. Aerodynamic priors on the same 110 noisy stable-spin entries. The two hit columns use each prior's own pooled 90% cutoff and the nominal prior's cutoff (0.8594). Areas use each prior's own cutoff. Rank one denotes the most probable goal cell.

Prior	Hits out of 110		Mean area (m^2)	Mean error (m)	Mean rank of true cell
	Own cutoff	Nominal cutoff			
Nominal	90	90	1.019	0.421	11.92
Half-widths $\times 0.5$	94	99	0.579	0.379	6.31
Half-widths $\times 1.5$	82	68	1.315	0.452	15.25
Higher S/E weights	101	94	1.592	0.405	11.65

In the post hoc threshold exchange, the 55 pooled Z09 calibration entries gave cutoffs of 0.8214 for aerodynamic prefix mass and 0.0800 for the gravity distance score. The 180 pooled Z10 entries gave 0.8594 and 0.0687, respectively. The aerodynamic cutoff rose, admitting more cells, and the gravity radius shrank. On the original 23 stable-spin entries the counts changed from 12 and 21 to 14 and 19. On the 110 newer entries the threshold source alone determined which model covered more kicks (Sec. 6.6).

Appendix I Reproducing the forecasts

The companion research materials contain the force laws, observations, per-kick predictions and reference crossings. The prescribed calculation uses seeds 8111 and 8117 on the 30×10 goal grid. The two fixed-deadline tests each provide separate calibration and test files, the original thresholds and individual scores. The report's reproduction guide gives commands for rebuilding the prescribed forecast, the rounded-coordinate witnesses and the timing and spin diagnostics. The frozen arrays suffice to recompute the figures and coverage tables without regenerating the kicks.

References

[1] University Physics Competition, 2026 contest, Problem B. Official problem statement.

[2] J. E. Goff and M. J. Carré, Soccer ball lift coefficients via trajectory analysis. *European Journal of Physics* **31**, 775–784 (2010). DOI 10.1088/0143-0807/31/4/007.

[3] T. Asai, K. Seo, O. Kobayashi and R. Sakashita, Fundamental aerodynamics of the soccer ball. *Sports Engineering* **10** (2007). DOI 10.1007/BF02844207.

[4] J. E. Goff, J. Kelley, C. Hobson, K. Seo, T. Asai and S. Choppin, Creating drag and lift curves from soccer trajectories. *European Journal of Physics* **38**, 044003 (2017). DOI 10.1088/1361-6404/aa6fcd.

[5] P. Virtanen et al., SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* **17**, 261–272 (2020). DOI 10.1038/s41592-019-0686-2.

[6] L. F. Shampine and M. W. Reichelt, The MATLAB ODE Suite. *SIAM Journal on Scientific Computing* **18**, 1–22 (1997). DOI 10.1137/S1064827594276424.

[7] T. Mizota et al., The strange flight behaviour of slowly spinning soccer balls. *Scientific Reports* **3**, 1871 (2013). DOI 10.1038/srep01871.

[8] International Football Association Board, Laws of the Game, Law 2: The Ball. Ball specifications, accessed 7 November 2026.

[9] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani and L. Wasserman, Distribution-Free Predictive Inference for Regression. arXiv:1604.04173.

[10] M. Bassek, R. Rein, H. Weber and D. Memmert, An integrated dataset of spatiotemporal and event data in elite soccer. *Scientific Data* **12**, 195 (2025). DOI 10.1038/s41597-025-04505-y. IDSSE v1 data, CC BY 4.0, DOI 10.6084/m9.figshare.28196177.v1.

[11] M. Gutiérrez-Pérez and A. Agudo, SoccerNet-v3D: Leveraging sports broadcast replays for 3D scene understanding. *IEEE/CVF CVPR Workshops* (2025). DOI 10.1109/CVPRW67362.2025.00595. ISSIA-3D annotations, release 1.0.0.

[12] I. Griffiths, C. Evans and N. Griffiths, Tracking the flight of a spinning football in three dimensions. *Measurement Science and Technology* **16**, 2056–2065 (2005). DOI 10.1088/0957-0233/16/10/022.

Report on the Use of AI

OpenAI Codex assisted with the physical analysis, numerical implementation, evidence checks, figure generation and revision of the English manuscript. This public draft contains the disclosure only. The complete original prompts and outputs are to be appended to the submission copy from the team's interaction records.